

CHEMISTRY | UNIT 03 | LESSON 03

Covalent Bonding and Molecular Compounds

Checkpoint Answer Sheet

Name: _____ Date: _____

MLA standards: MLA.CHEM.BON.01 and MLA.CHEM.BON.02

Complete independently. Show electron sharing, formula meaning, and evidence-based reasoning.

1. Draw and label a Lewis model for H₂O. Show both shared electron pairs and the two lone pairs on oxygen.

2. Explain why H₂O is a molecule rather than an ionic formula unit.

3. CO₂ contains one carbon atom and two oxygen atoms. Explain how its formula and bonding model communicate the same information.

4. Correct this misconception: A molecular formula can always be reduced to the lowest whole-number ratio.

5. Use claim-evidence-reasoning to explain why atoms share electrons in a covalent bond.